
An LLM Wiki System for High-Consequence Domain Knowledge Organization

Initial Feasibility for Space Operations and Nuclear Engineering

Evaluated on open technical literature

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Outline

- | | | |
|---|---------------------|--|
| 1 | Motivation | Why build this |
| 2 | The Wiki | A model that writes, organizes, and reads back its own notes — and why that is not standard RAG |
| 3 | Architecture | Write → Organize → Read, shown on the running system |
| 4 | Evaluation | What we measured on the wiki — and what we did not |
| 5 | Steering | The Idea Compass — patent-pending — and where the claim ends |

1 Motivation

A **wiki** is pages people keep · **RAG** searches document pieces when asked · **an LLM wiki** is pages the model writes, and re-reads itself

"Instead of just retrieving from raw documents at query time, the LLM incrementally builds and maintains a persistent wiki."

Andrej Karpathy

"LLM Wiki" · gist.github.com · 4 Apr 2026

THE PROBLEM

Session 1



lost

Session 2



lost

Session 3

"...the LLM retrieves relevant chunks at query time, and **nothing accumulates.**" A maintained wiki keeps what was built — and reads it back **without being asked.**

OPEN CONCEPT · PRIOR ART

the LLM-maintained wiki — published by Karpathy, Apr 2026

OUR CONTRIBUTION

our implementation of it — and a feasibility study

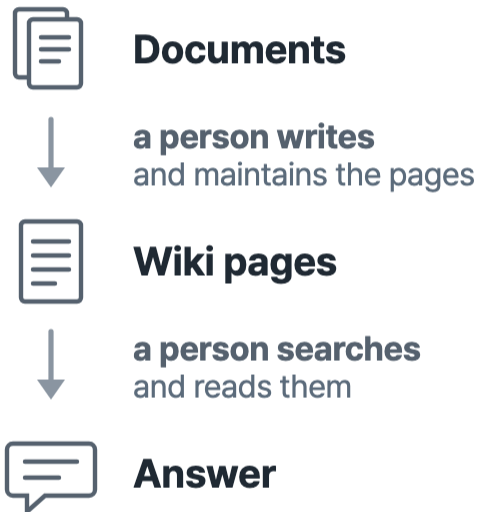
PATENT-PENDING · KAERI

the Idea Compass steering mechanism only

2 The Wiki

Traditional Wiki

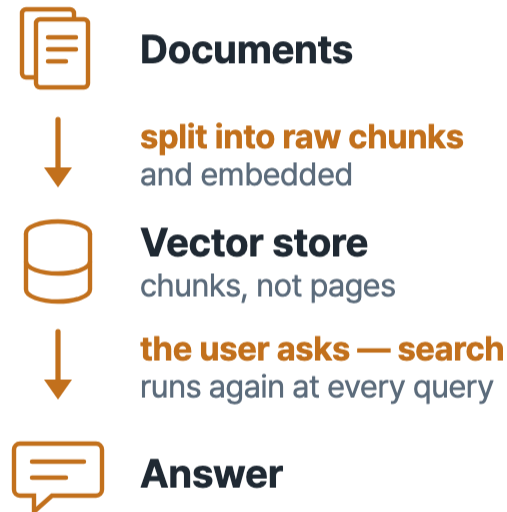
Pages that people write by hand, and keep up to date themselves.



Compiled once — but by people, so it goes stale

RAG

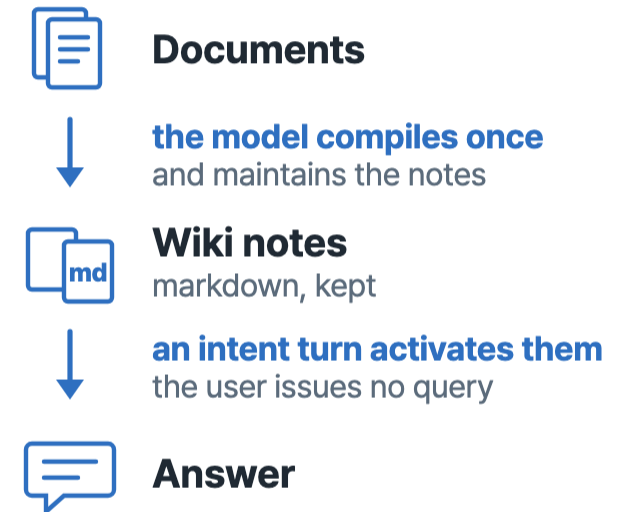
Retrieval-augmented generation. No pages are kept — the store is searched every time.



No maintained pages — the chunks are re-searched every query

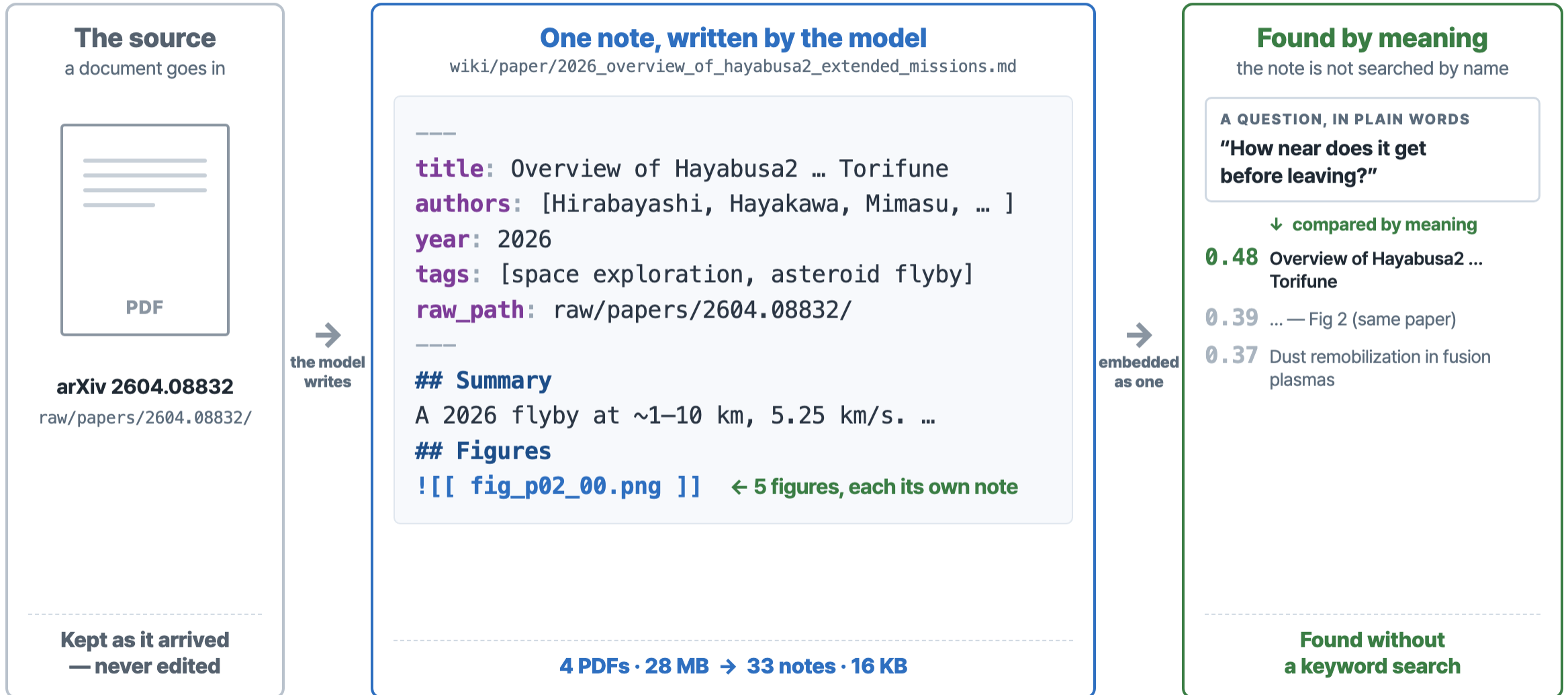
LLM Wiki

The model writes the pages, and keeps them up to date, itself.

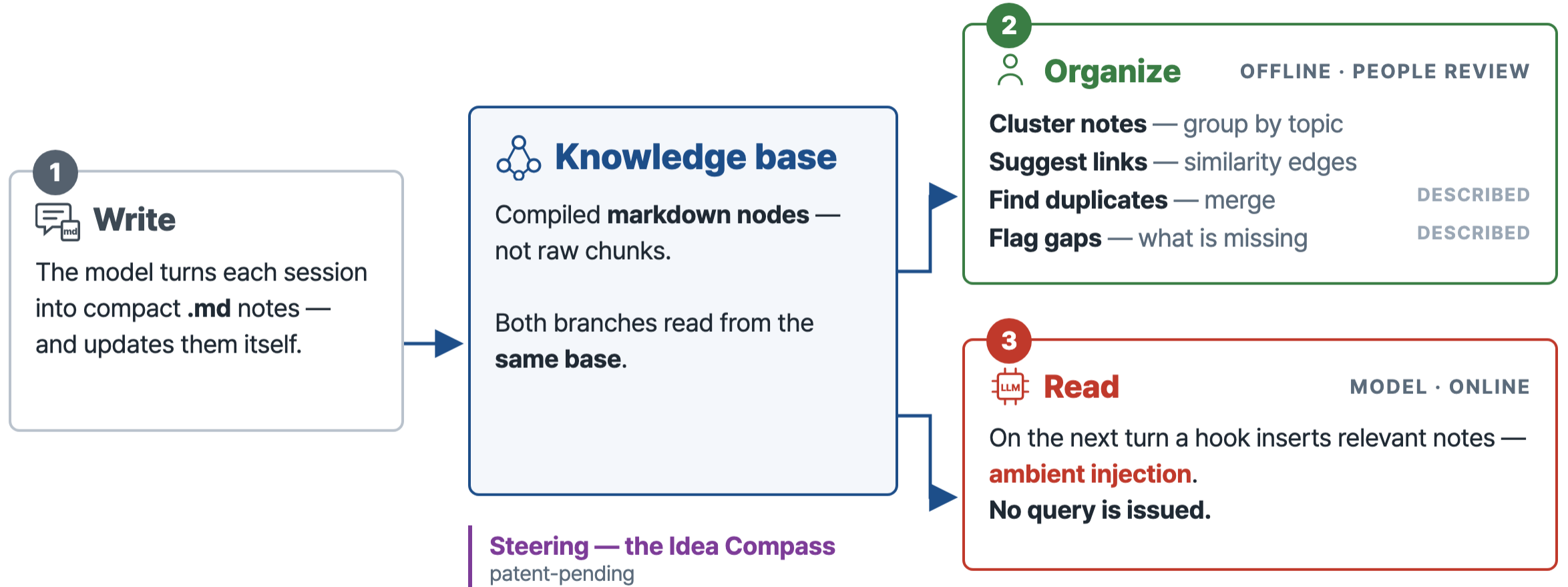


Compiled once and kept — the upkeep moves to the model

2 The Wiki



3 Architecture



Research Universe

155 public arXiv papers · the wiki organizes them itself — abstract embedding → clusters (space · AI · nuclear) with similarity links

● Space ● AI / LLM ● Nuclear

◆ similarity links · on

arrange by

Domain

practical ↔ novel

hardware ↔ software

incremental ↔ breakthrough

NUCLEAR

SPACE

AI / LLM

155 papers · 553 links

Public arXiv only · self-organized

1 · Organize

2 · Notice

3 · Idea Compass ◆

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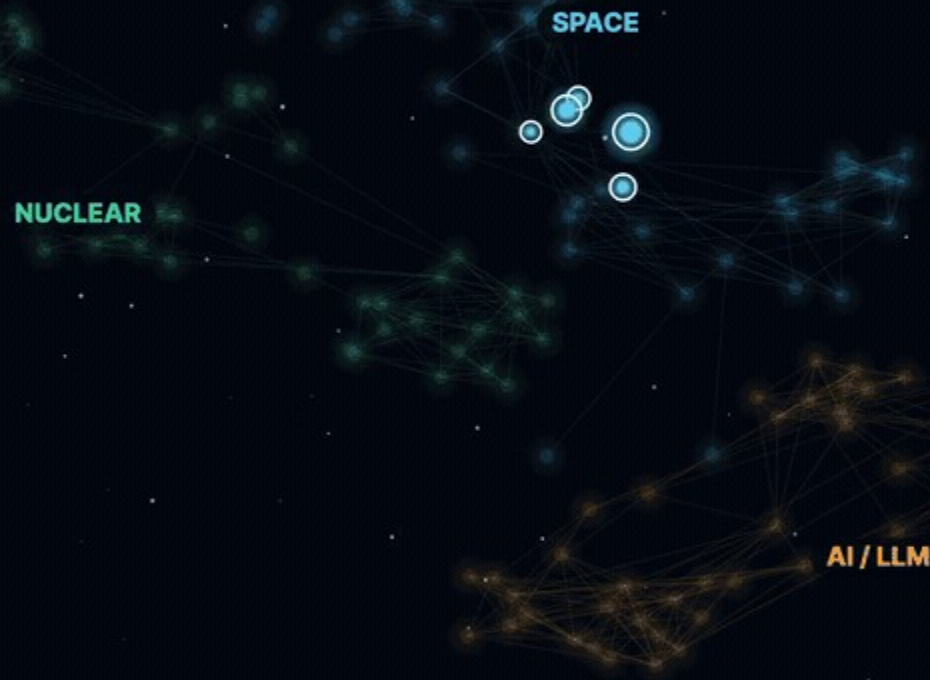
Research Universe

It notices what you are doing — an intent turn (a plan, not a question) auto-retrieves the relevant papers, no search

● Space ● AI / LLM ● Nuclear

INTENT TURN · not a question

"let us check the satellite power margin before the eclipse and plan the next downlink pass"



◆ LLM-compiled wiki summaries · 5

not raw chunks — each is a maintained .md node

Multidisciplinary Design Optimization Ap .md ↗

Space mission planning and spacecraft design are tightly coupled a...

▸ matched: spacecraft

Spacecraft Fiducial Marker for Autonomou .md ↗

Robotic operations in space are challenging due to the harsh envir...

▸ matched: spacecraft

Operations for Autonomous Spacecraft .md ↗

Onboard autonomy technologies such as planning and scheduling, ide...

▸ matched: spacecraft

+ 2 more summaries

1 · Organize

2 · Notice

3 · Idea Compass ↗

3 Before / After

ONE QUESTION

A satellite power margin before an eclipse

same question · same model — only the context differs

 BEFORE

No context — closed-book

nothing injected

"I can't determine whether the eclipse pass is survivable from this alone. ... I'd need the eclipse duration, the average load, battery capacity, and the allowed DoD."

→ lists what it would need

EXCERPT

 AFTER

With the wiki's context — ambient

 **wiki notes injected — no query**

"End the high-power observation **before eclipse entry**, then switch to low-power standby. ... Keep the projected end-of-eclipse depth-of-discharge **within the operational limit**."

→ uses the noted constraints

EXCERPT

4 Evaluation

Preliminary, single-rater observations. Reported as direction, not as validated results.

- On intent turns, a **naive** query-triggered baseline did not activate the relevant notes
it waits for a question — an intent turn is not phrased as one. A fair baseline is future work.
- In a small pairwise comparison, **one** human rater preferred the ambient-context condition
the rater judged which context felt more useful — not which answer was factually correct.
- The LLM judge and the human agreed only weakly

AMBIENT — TURNS FIRED

40 / 40

cumulative session context

CURRENT TURN ONLY

40 / 40

same retriever, context removed

QUESTION-TRIGGERED

0 / 40

fires only when the turn is a question

4 Results

The paper's controlled experiment — a ~6,000-note wiki, 56 answerable questions.

RETRIEVAL — RECALL@3



ANSWER CORRECTNESS



5 Steering

The Idea Compass

PATENT-PENDING

- This is the part we claim
- What it does: **coarse directional steering** of idea generation
- Because it is patent-pending, the internal mechanism is not explained here

demo follows →

Set your axes and the map re-arranges to your frame — pick anchors, steer toward the gap, then

Idea Compass

coarse directional

steered idea generation · you set the bearing, it proposes

- 1 Set your axes — the map re-arranges to your frame

practical ↔ novel × applied ↔ foundation: ▾

- 2 Pick 2–3 anchor papers (0)

- 3 Steer the bearing arrow toward the gap



practical novel

- 4 Reach — how far into the frontier

near

·

far

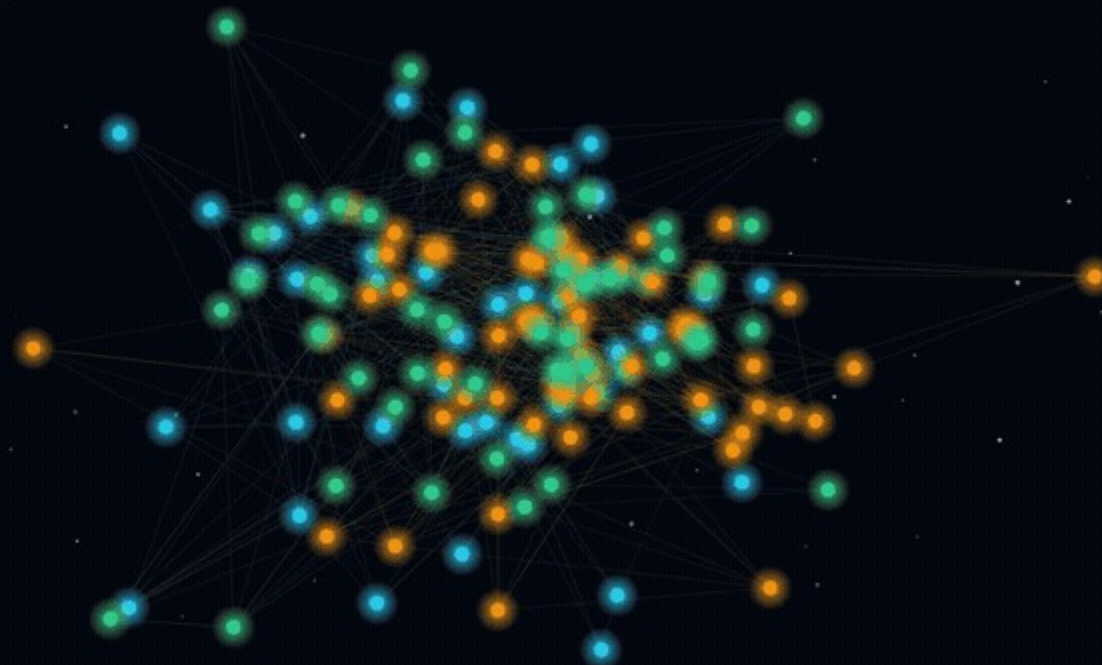
close idea

bolder leap

↶ Flip bearing (sign-flip)

Generate

Change the axes and the layout changes — different frame, different gaps. Positions are approximate (not measured coordinates); the idea lands **near** the bearing.



◀ practical

novel ▶

1 · Organize

2 · Notice

3 · Idea Compass ↕

Set your axes and the map re-arranges to your frame — pick anchors, steer toward the gap, then

Idea Compass

coarse directional

steered idea generation · you set the bearing, it proposes

- 1 Set your axes — the map re-arranges to your frame

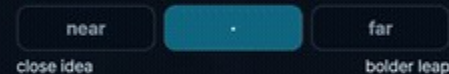
hardware ↔ software × incremental ↔ bold

- 2 Pick 2–3 anchor papers (3)

- 3 Steer the bearing arrow toward the gap



- 4 Reach — how far into the frontier



◁ Flip bearing (sign-flip)

Generate

Change the axes and the layout changes — different frame, different gaps. Positions are approximate (not measured coordinates); the idea lands **near** the bearing.

incr

◀ hardware

software ▶

1 · Organize

2 · Notice

3 · Idea Compass

IDEA COMPASS · SPACE · illustrative

bearing: toward hardware / software

Spacecraft Operations Hardware-Software Testbed

hardware software

Grounded on 3 public papers: arXiv 2111.06372v1 · arXiv 2604.08832v1 · arXiv physics/0508019v1
Generated offline from these 155 public arXiv abstracts — the five stops were produced together in one pass, not independently controlled, and not live. Lands near the bearing, not on a coordinate. Mechanism patent-pending.

~ near this bearing

5 Limitations

What this study shows — and what it does not.

DEMONSTRATED

Query-less context activation on intent turns

a running prototype that writes, organizes, and reads its own markdown wiki

OBSERVED

Coarse directional steering — the Idea Compass

patent-pending mechanism · behavior shown, not validated

NOT ESTABLISHED

Superiority over stronger RAG baselines · idea quality · precise targeting

we did not build a query-rewrite, intent-detection, or agentic baseline — and judge–human agreement was weak

In short — the human pilot was one rater, small sample.

6 Conclusions

The model writes its own wiki, organizes it, and reads it back — without being asked.

TAKE THIS WITH YOU

On an intent turn, relevant context is activated. **No query is issued.**

A patent-pending component adds coarse directional steering.

LEAVE THIS HERE

I make no claim about idea quality.

No superiority over RAG baselines was shown.

The **human-rated** evidence is preliminary and single-rater.

If this direction holds up, the next step is a real operations setting — the setting this system was built for.

Thank you.

Juhyeok Park

complimentary656@gmail.com · Applied AI Section, KAERI

Questions welcome.

Backup slides

Prepared material. Slide numbers on the right.

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A1 Is this just RAG?

BACKUP

LIKELY Q “Query-less activation is just RAG with extra steps, isn't it?”

- **Standard RAG needs an explicit query** → it fetches raw chunks on demand

On an **intent turn** — a plan, not a question — no query is issued at all.

- **Here, compiled .md summaries activate** — not raw chunks

maintained nodes, injected as ambient context.

- **This is a design choice we show** — not a claim to beat RAG

Motivating audit: one query-triggered baseline fired on **0 of 40** intent turns. One baseline, one audit.

We did **not** build a query-rewrite or agentic-RAG baseline — superiority over those is **not** claimed.

A2 Evaluation — what we measured

BACKUP

LIKELY Q “How do you actually evaluate this?”

- **Preliminary, single-rater — reported as direction**
not a population estimate; a second rater is future work.
- **Query-less activation on intent turns — demonstrated on the running system**
the baseline did not activate on the audited intent turns.
- **Small pairwise preference — one human rater preferred the ambient answers**
small sample ($N \approx 16$). The word “one” matters.

Exact per-condition figures are in the **paper**. Interpretation here is deliberately conservative: directional, not validated.

A3 Judge reliability

BACKUP

LIKELY Q “You used an LLM judge — how reliable is it?”

- **Agreement between the LLM judge and the human was weak**

we do not lean on the automated score.

- **We treat this as a limitation, not a finding**

it is why we frame everything as directional.

- **Single rater → interpretation is limited**

a second human rater is the clear next step.

We report the weak agreement **openly** rather than hide it — honesty over a stronger-sounding claim.

A4 The steering mechanism

BACKUP · PATENT-PENDING

LIKELY Q “How does the Idea Compass actually generate the idea?”

- **We show existence and observed behavior only**

pick a lens & anchors → set a coarse bearing → an idea lands near that region.

- **Coarse direction — not precise targeting**

positions are approximate; we make no idea-quality claim.

- **The generation mechanism is patent-pending**

it is not shown or validated here; the demo clips are pre-recorded.

| **What is claimed:** a coarse directional steering behavior. **What is not:** idea quality, precise coordinates, or a validated generator.

A5 System & methods

BACKUP

LIKELY Q “What's under the hood — models, corpus, pipeline?”

PIPELINE

- **Write**
model turns each session into compact .md notes
- **Organize · people · offline**
cluster · find duplicates · suggest links · flag gaps
- **Read · model · online**
a hook inserts relevant notes — ambient injection, no query

METHODS & DATA

- **Embedding**
abstracts embedded with bge-m3
- **Projection**
UMAP → 2-D map; clusters emerge on their own
- **Corpus**
155 public arXiv papers, collected by topic search around three themes — space · AI · nuclear (illustrative)

A6 Contribution & domain

BACKUP

LIKELY Q “If the wiki idea is Karpathy's, what did you do?”

- **The concept is credited to Andrej Karpathy**

an LLM that writes and maintains its own structured wiki, as an alternative to query-time RAG.

- **Our contribution: from a personal tool to a studied system**

we implement the full pipeline (write · organize · read) and measure feasibility.

- **Why space & nuclear**

critical knowledge must be on hand, yet is rarely queried in the moment — where query-less context could matter.

We add a **patent-pending** steering component on top — coarse directional only, quality not claimed.

A7 How “query-less” actually works

BACKUP

LIKELY Q “If there's no query, what selects the context — isn't this just retrieval?”

- An **intent turn** is a plan the user types — not a search string
the user never authors a query and there is no explicit query-rewrite step.
- Internal relevance scoring **does** remain
the turn is embedded and scored against the maintained .md nodes; top nodes are selected, deduplicated, and compiled into the prompt.
- “Query-less” = **no user query** — not retrieval-free
we do not claim to remove similarity scoring; we claim the user issues no search.

A8 The baseline was naive — on purpose acknowledged

BACKUP

LIKELY Q “Did you rig the baseline? 0 of 40 sounds too clean.”

- The audited baseline was a **naive query-triggered lookup**

it waits for a question; an intent turn is not phrased as one, so it did not fire.

- This is **not** “RAG fails”

it is one naive baseline, one small audit — we say so on the slide.

- A **fair** comparison is future work

query-rewrite, intent-classifier, and conversation-aware retrieval — same corpus, embedding, prompt & token budget.

We do **not** claim superiority over query-rewrite or agentic RAG. Building those baselines is the honest next step.

A9 Where it fails

BACKUP

LIKELY Q “You only showed successes — what breaks?”

- **Organize** — a paper can land between clusters
embedding similarity is imperfect; some nodes sit ambiguously.
- **Notice** — activation can miss a relevant node, or pull an off-topic one
selection is scored, not perfect; recall is not guaranteed.
- **Compass** — a bearing can point at a dense region, not a sparse one
the direction is coarse and approximate — never a precise coordinate.

Success-only demos are a limitation. The clips are best-runs on a public corpus, not a claim of reliability.

A10 Scale, latency, cost

BACKUP

LIKELY Q “155 nodes is a toy — does this survive a real wiki?”

- Demonstrated at **155 nodes** — a small, illustrative corpus
we make no scale claim beyond this.
- **Organize / write** run offline; **read** adds a lightweight hook
the maintenance work is human-facing and not on the response path.
- At larger scale, the open risks are **context saturation and noise**
as more nodes activate, prompt budget and relevance both get harder — not yet benchmarked.

Latency, write/update cost, and behavior beyond a few hundred nodes are **not measured**. That is future engineering, not a current claim.

A11 Safety & domain boundary

BACKUP

LIKELY Q “This is space & nuclear — is any of it operationally usable?”

- **Public research corpus only**

no operational, restricted, proprietary, or export-controlled data was used.

- **Not a decision-support system**

the eclipse example is illustrative and LLM-generated — not validated operating guidance.

- **A human gate is required before any real use**

provenance, access control, and verification would all be prerequisites we have not built.

Nothing here is safety-validated. Domain is the motivation; the evidence is a feasibility signal on open literature.

A12 Patent boundary

BACKUP · LEGAL

LIKELY Q “What exactly is patented, and does this talk prove novelty?”

- **Patent-pending** — filed, **not granted**

this talk does not claim to prove novelty, patentability, or commercial readiness.

- The **wiki pipeline is not claimed** — the concept is Karpathy's

only the **Idea Compass steering mechanism** is the patent-pending contribution (KAERI).

- The mechanism is **intentionally not fully disclosed here**

we show existence and behavior; the generator itself is out of scope for this talk.

Claim boundary in one line: **open** = the wiki concept · **ours** = the pipeline + study · **patent-pending** = the steering mechanism only.

A13 Related work, and what we claim

BACKUP

LIKELY Q “How is this different from existing agent-memory or graph-RAG systems?”

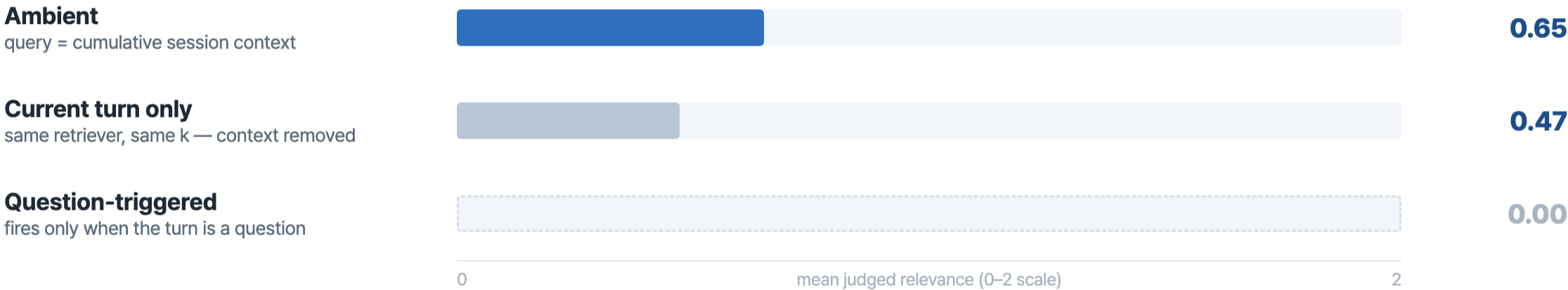
- **MemGPT** (Packer et al., 2023) — the model pages its own context window
ours: offline-organized markdown nodes, injected ambiently on intent turns.
- **Generative Agents** (Park et al., 2023) — memory stream with reflection
ours: an explicit write · organize · read pipeline, with offline human curation.
- **Reflexion** (Shinn et al., 2023) — self-reflection to retry one task
ours: a persistent cross-session knowledge base, read without being asked.
- **GraphRAG** (Edge et al., 2024) — LLM-built graph, answered at query time
ours: the read path carries no user query.
- **Linear analogies** (Mikolov et al., 2013) — direction arithmetic on embeddings
the patent covers the steering mechanism, not linear arithmetic.

We do not claim memory, summarization, or embedding arithmetic as new. **Claimed** = the pipeline as built and studied · **patent-pending** = the Idea Compass mechanism only.

A14 What each condition actually retrieved

BACKUP

LIKELY Q “Zero out of forty? Did you rig the baseline?”



AMBIENT — TURNS FIRED

40 / 40

CURRENT TURN — TURNS FIRED

40 / 40

QUESTION-TRIGGERED — TURNS FIRED

0 / 40

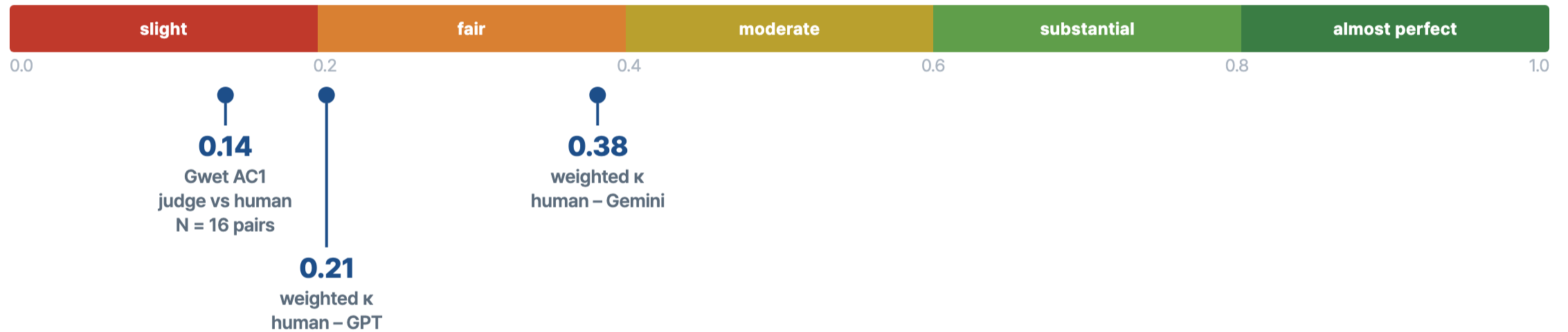
N = 40 turns · 170 nodes judged · single rater · blind to condition · relevance 0–2

All 40 turns state an intent rather than ask — and **intent turns are 28% of 7,547 logged turns**, not a rare case. The 0/40 is structural, not a score: a question-triggered baseline never fires, so there is nothing to grade. It is **one naive baseline, by design** — we did not build a query-rewrite or agentic baseline.

A15 How far do LLM judges agree with the human?

BACKUP

LIKELY Q “You say agreement was weak — weak by what measure?”



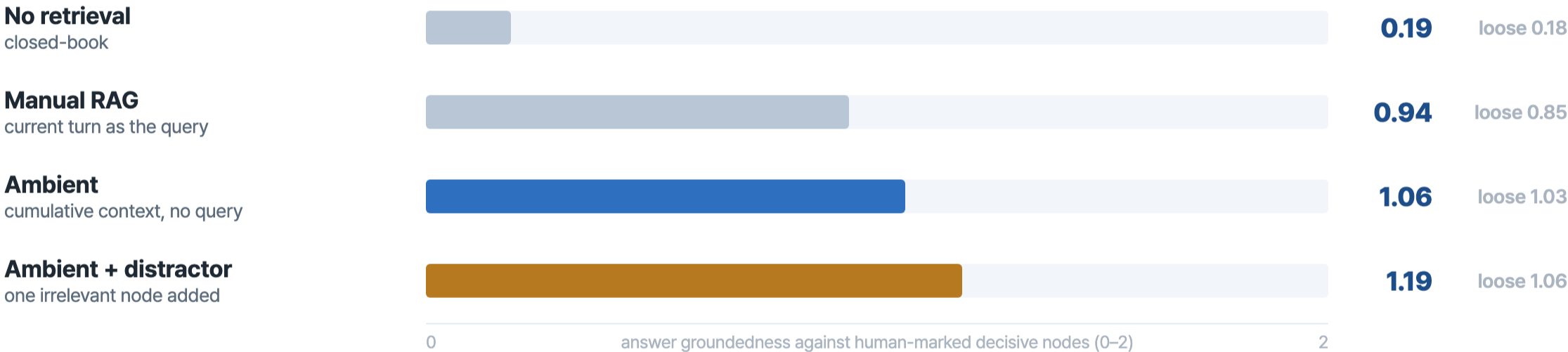
n = 170 node judgments (κ) · N = 16 pairs (AC1) · single human rater · Landis-Koch bands

Every measure lands in **slight to fair**. Same-family and cross-family judges scored the **same** (AC1 0.14 both), so swapping model families does not rescue it. With one rater we cannot separate “the judge differs from people” from “this rater is noisy” — **a second rater is the next step**. We report this as a limitation, not as a finding about the field.

A16 Does the retrieved context reach the answer?

BACKUP

LIKELY Q “Fine, it retrieves — but does that change the answer at all?”



N = 16 turns strict / 34 loose · generator GPT-5.6 · single judge Gemini-3.6 · gold = nodes a human marked decisive

Without any context the answer collapses (0.19); in the conditions we tested, retrieval raised it. **The distractor row is not a result we claim** — at N = 16 we read 1.06 vs 1.19 as noise, not as evidence that irrelevant context helps. This measures **groundedness, not factual accuracy**, and one judge scored it.

A17 Is corpus size the bottleneck?

BACKUP

LIKELY Q “This works on 155 notes. What happens when the wiki gets big?”

HIT RATE AS THE CORPUS GROWS — 50 → 500



flat from 123 onward · 4 corpus sizes × 30 questions each

HIT RATE BY QUESTION TYPE — ALL SIZES POOLED



compositional = the answer needs two or more notes joined

nested corpora 50 < 123 < 300 < 500 public abstracts · 30 pre-registered questions per size · 360 retrievals

Growing the corpus tenfold did not move the hybrid retriever. **What varied was the question type, not the size** — when an answer needs two notes joined, we hit it half the time. 500 is still small, so this says nothing about a plant-scale corpus; it says size was not what limited us here.

A18 Which corpus, and why it is the easy case

BACKUP

LIKELY Q “Abstracts are already summaries. Isn’t that the easy case?”

THREE CORPORA — ALL PUBLIC ARXIV

Demo on screen

155

Research Universe · space · AI · nuclear

Domain experiment

123

space 63 · nuclear 60

Retrieval study

500

nested 50 $\subset$ 123 $\subset$ 300 $\subset$ 500

- Assembled by **topic search** around three themes. We do not claim every paper sits squarely in that field.
- No operational, restricted, proprietary or export-controlled text was used. Public literature only, by design.

the same corpus family throughout · abstracts only · no full-text run

WHY THIS IS THE EASY CASE

- An abstract is **already an author-written summary** — we never had to choose what matters inside 8,000 words.
- The input is **clean** — no tables, equations, captions or reference lists.
- One node is **one topic** — “which part of this document is relevant” never comes up.
- So retrieval **looks better here** than it would on full documents.

Full documents are the real test, and we have not done it. The evaluation is measured on this corpus, so changing the corpus means redoing the evaluation — that is future work, not a quick fix.

A19 What runs where

BACKUP

LIKELY Q “Could this run offline, on-premise, or on a closed network?”

HOSTED APIS TODAY

- Generation — GPT-5.6
- Judging — Gemini-3.6
- Both are external services. Nothing in this study required them to be, but that is how it was run.

LOCAL BY CONSTRUCTION

- The wiki itself — plain markdown files on disk.
- Embedding — bge-m3, open weights.
- Clustering and the 2-D map — computed from those embeddings.
- Text only. Diagrams, tables and telemetry are out of scope.

no air-gapped or on-device run was attempted · no latency or cost measured under such a setting

We have not tested an offline or closed-network deployment, so we claim nothing about it. The parts that would have to be replaced are the hosted generation and judging calls; the store, the embedding model and the map are already local.

A20 Who does the work

BACKUP

LIKELY Q “How much human effort does Organize need — doesn’t that limit it?”

STEP	WHO	WHEN	SHOWN TODAY
Write — session → markdown notes	model	per session	demonstrated
Organize — cluster, suggest links	system	offline	demonstrated
Organize — merge duplicates, flag gaps	system	offline	described, not demonstrated
Curate — accept, correct, discard	human	offline	not measured
Read — ambient injection	model	online, every turn	demonstrated

Organize and Write sit off the response path — they are maintenance, not latency

We did not measure human curation time, so we make no efficiency claim. The honest position is that a person stays in the loop for Organize, and whether that stays affordable at a larger wiki is exactly what we have not tested.

A21 How the paper experiment was graded

BACKUP

LIKELY Q “Two language models agreeing with each other — why should I trust that?”

WHAT WAS ASKED

- A working research wiki — about 6,000 notes across eight categories.
- 56 questions that **have an answer** in it.
- 16 more where the correct behaviour is **to refuse**.
- For each one, **a person marked the exact passage that answers it**.

HOW IT WAS GRADED

- The judge **could not see** which method produced the answer.
- The judge checked one thing — **does the answer contain that marked passage?**
- Two language models did this, from **different families**.
- They gave the same label 94% of the time — raw agreement, not chance-corrected.

We asked the judge a smaller question on purpose. Not “is this answer good” — “does it contain the passage a person marked”. That is why the number means something. It still only says the labels were reproducible, not that they were right; human adjudication is future work.